Keys and Constraints

Course: MIT 8103 Advanced Database Systems
Portfolio: 1 — Database Design and Modelling
Case Study: Student Academic Management and Result Processing System (SAMRPS)
Database: sarms_db
DBMS: MySQL/MariaDB

1. Introduction

Keys and integrity constraints are essential components of any relational database. The sarms_db for example follows this design approach. They identify records uniquely, preserve relationships between tables, prevent duplicate data and enforce valid academic information.

The sarms_db uses primary keys, foreign keys, unique constraints, domain restrictions and check constraints.

2. Primary Keys

A primary key uniquely identifies each record in a table. Primary keys must be unique and cannot contain NULL values.

Table	**Primary key**
departments	department_id
programmes	programme_id
students	student_id
lecturers	lecturer_id
courses	course_id
academic_sessions	session_id
semesters	semester_id
course_offerings	offering_id
course_registrations	registration_id
results	result_id
users	user_id
audit_logs	audit_id

These keys enforce entity integrity by ensuring that each row has a unique identifier.

3. Foreign Keys

Foreign keys link related tables and enforce referential integrity.

Table	Foreign key	Referenced table
programmes	department_id	departments.department_id
students	programme_id	programmes.programme_id
lecturers	department_id	departments.department_id
courses	department_id	departments.department_id
semesters	session_id	academic_sessions.session_id
course_offerings	course_id	courses.course_id
course_offerings	semester_id	semesters.semester_id
course_offerings	lecturer_id	lecturers.lecturer_id
course_registrations	student_id	students.student_id
course_registrations	offering_id	course_offerings.offering_id
results	registration_id	course_registrations.registration_id
audit_logs	user_id	users.user_id

For example, a student must reference an existing programme, and a result must reference an existing course registration.

4. Unique and Candidate Keys

Unique constraints prevent duplicate values in attributes that identify business records.

Table	Unique attribute(s)	Purpose
departments	department_code, department_name	Prevent duplicate department codes and names
programmes	programme_code	Ensure programme codes are unique
students	matric_no, email	Prevent duplicate matriculation numbers and email addresses
lecturers	staff_no, email	Prevent duplicate staff numbers and email addresses
courses	course_code	Ensure course codes are unique
academic_sessions	session_name	Ensure session names are unique
semesters	(session_id, semester_name)	Prevent duplicate semester names within a session

course_offerings	(course_id, semester_id)	Prevent duplicate offerings of the same course in a semester
course_registrations	(student_id, offering_id)	Prevent duplicate student registrations for an offering
results	registration_id	Permit at most one result record per registration
users	username	Prevent duplicate usernames

A candidate key is a minimal attribute or combination of attributes that can uniquely identify a record. The primary key is the candidate key selected as the table's main identifier. Other unique constraints may enforce additional candidate keys where the attributes satisfy the required conditions.

5. Domain and Check Constraints

The design includes the following data rules:

- **Programme duration:** duration_years must be between 1 and 5.
- **Course credit units:** credit_unit must be between 1 and 6.
- **Academic session dates:** end_date must be later than start_date.
- **Continuous assessment score:** ca_score must be between 0 and 40.
- **Examination score:** exam_score must be between 0 and 60.
- **Grade point:** where provided, grade_point must be between 0 and 4.
- **Total score:** total_score is derived from ca_score + exam_score.
- **Academic level:** student levels are restricted to the defined ND and HND levels.
- **Status and role fields:** controlled values are used for student, registration, result, session and user statuses and roles.

The NOT NULL constraints in the DB ensure that required fields cannot be left empty. UNIQUE constraints protect against duplicate business identifiers.

6. Integrity Rules and Their Importance

Entity integrity: Primary keys identify records uniquely and cannot be NULL.

Referential integrity: Foreign keys ensured that references to departments, programmes, students, lecturers, courses, semesters, offerings, registrations and users point to valid records.

Domain integrity: Data types, allowed values and check constraints restrict values to valid ranges and categories.

Business-rule integrity: Unique combinations prevent duplicate course offerings and duplicate student registrations. The result design permits at most one result record per registration.

Audit integrity: Audit log records was linked to a user account so that system activity can be associated with an actor.

7. Database Table Definition

The following commands show the actual table definition:

SHOW CREATE TABLE departments;

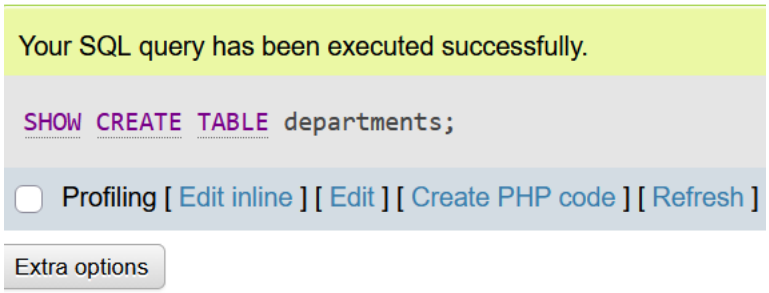


Table	Create Table
departments	CREATE TABLE `departments` (`department_id` int...

SHOW CREATE TABLE programmes;



Table	Create Table
programmes	CREATE TABLE `programmes` (`programme_id` int(1...

SHOW CREATE TABLE students;

Your SQL query has been executed successfully.

```
SHOW CREATE TABLE students;
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Table	Create Table
students	CREATE TABLE `students` (`student_id` int(10) u...

SHOW CREATE TABLE lecturers;

```
SHOW CREATE TABLE lecturers;
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Table	Create Table
lecturers	CREATE TABLE `lecturers` (`lecturer_id` int(10)...

SHOW CREATE TABLE courses;

```
SHOW CREATE TABLE courses;
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Table	Create Table
courses	CREATE TABLE `courses` (`course_id` int(10) uns...

SHOW CREATE TABLE academic_sessions;

```
SHOW CREATE TABLE academic_sessions;
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Table	Create Table
academic_sessions	CREATE TABLE `academic_sessions` (`session_id` ...

SHOW CREATE TABLE semesters;

```
SHOW CREATE TABLE semesters;
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Table	Create Table
semesters	CREATE TABLE `semesters` (`semester_id` int(10)...

SHOW CREATE TABLE course_offerings;

```
SHOW CREATE TABLE course_offerings;
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Table	Create Table
course_offerings	CREATE TABLE `course_offerings` (`offering_id` ...

SHOW CREATE TABLE course_registrations;

```
SHOW CREATE TABLE course_registrations;
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Table	Create Table
course_registrations	CREATE TABLE `course_registrations` (`registrat...

SHOW CREATE TABLE results;

```
SHOW CREATE TABLE results;
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Table	Create Table
results	CREATE TABLE `results` (`result_id` int(10) uns...

SHOW CREATE TABLE users;

```
SHOW CREATE TABLE users;
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Table	Create Table
users	CREATE TABLE `users` (`user_id` int(10) unsigne...

SHOW CREATE TABLE audit_logs;

```
SHOW CREATE TABLE audit_logs;
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Table	Create Table
audit_logs	CREATE TABLE `audit_logs` (`audit_id` bigint(20...